radar, enhances accuracy and robustness in uncertain conditions. Recent advances in machine learning and AI-based navigation algorithms further improve real-time path estimation, enabling drones to recognize environmental patterns, adapt to obstacles, and predict movement trajectories without external guidance (Lu, et al. 2022).

In military and defense applications, where GPS denial may be intentional due to electronic warfare or signal jamming, adaptive strategies like terrain-relative navigation, radio-frequency mapping, and cooperative multidrone localization are being implemented. These techniques allow a swarm of drones to share positional data and collectively maintain accuracy, even when external references are unavailable.

The ability to achieve precise navigation in GPS-denied environments remains a cornerstone of fully autonomous drone operations. The integration of AI-driven perception, multi-sensor fusion, and vision-based mapping techniques is steadily enhancing positional reliability under complex and constrained conditions. As research advances, these technologies will enable drones to operate intelligently and self-sufficiently, ensuring accurate localization, safe maneuvering, and mission continuity even in environments where traditional satellite-based guidance systems are unavailable.

7.3.2 Developing reliable obstacle detection and avoidance systems

For autonomous drones to operate safely and efficiently, especially in dynamic or cluttered environments, the development of reliable obstacle detection and avoidance systems is paramount. This capability enables drones to perceive their surroundings, identify potential hazards, and execute real-time maneuvers to prevent collisions, a critical requirement for